

Data Mining / Big data

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Instructors

Aris (Aris Anagnostopoulos)



Luca (Luca Becchetti)



Who am I?

Professor in computer engineering at Sapienza

Studied in Greece, USA

Worked in the US for a bit, then came to Sapienza

Works on algorithms, data mining, data analysis, ...

Search engines, social networks, recommender systems,
misinformation, biology, finance, medicine, ...

Who are you?



Who am I (L. B.)?

Luca Becchetti

I work on algorithms and theory for data mining, machine learning and distributed processes

Recently: mining large (possibly temporal) graphs and datasets, algorithms for high-dimensional spaces, algorithmic aspects of complex systems

Always: probabilistic analysis

Who are you?



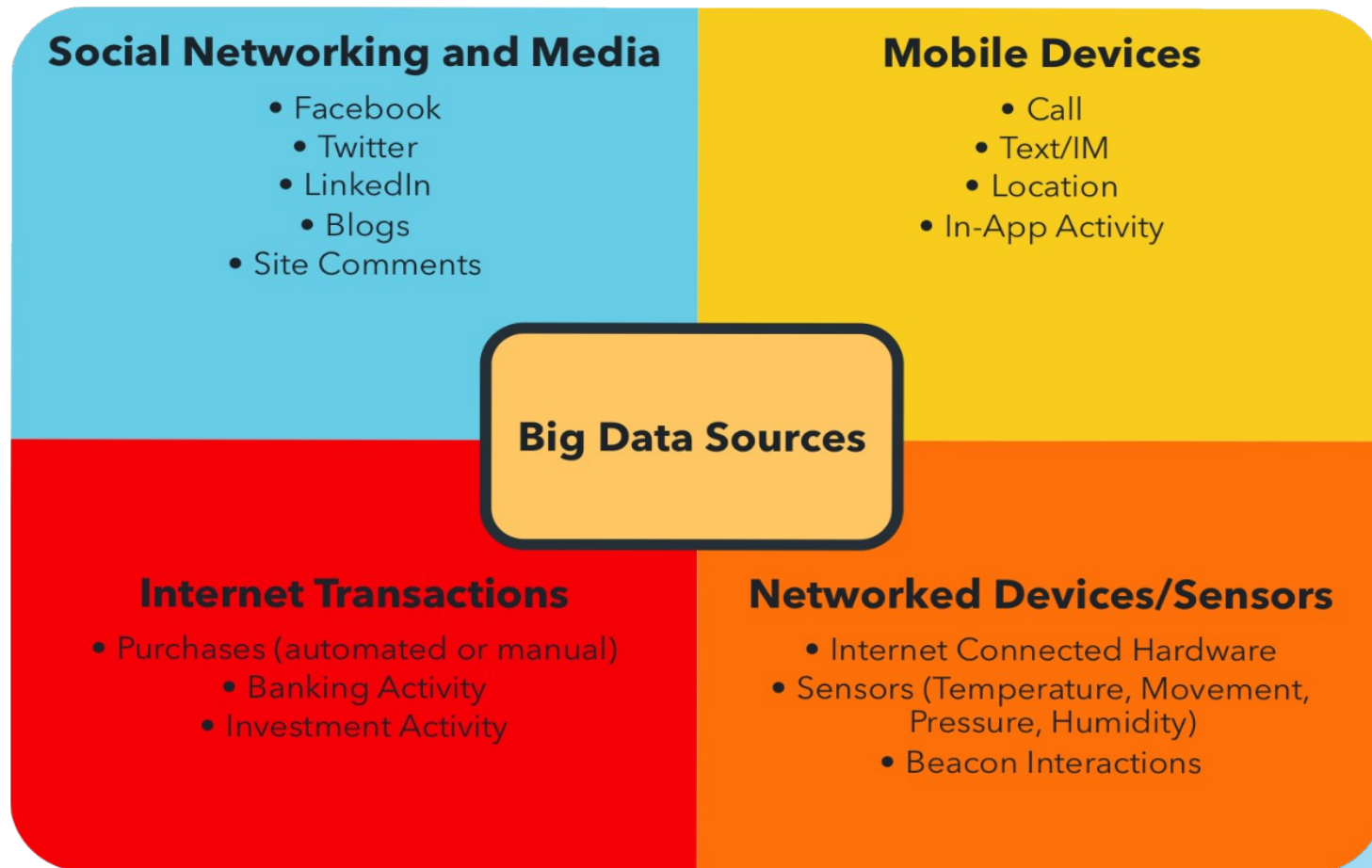
Goals

- Advanced tools used in data analysis that are currently not covered in the basic curriculum
 - Algorithms and intuitions/theory behind
 - Implementation in realistic scenarios
 - Emphasis is on datasets that are somehow “big”
 - More about meaning of “big” later in this lecture
- 

Where do data come from?



Data sources



Digital universe and digital footprint



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Your digital footprint
(or shadow)

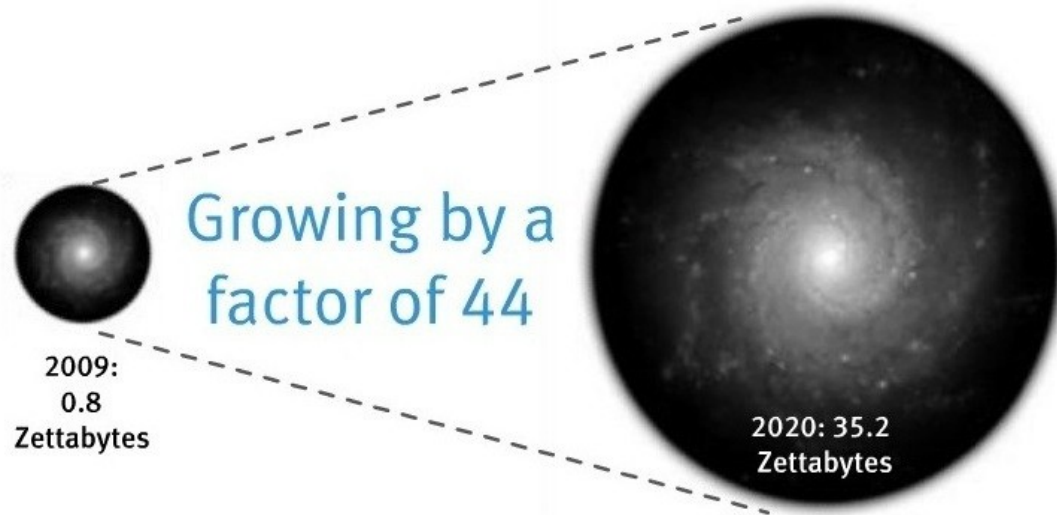


How big is “big”?

The growth of things

Source: IDC Digital Universe Study,

The Digital Universe 2009 to 2020



Equivalent to a stack of DVDs in 2009 reaching to the moon and back, now reaching halfway to Mars by 2020

Source: IDC Digital Universe Study

DATA DELUGE

The billions of terabytes (TB) produced in one year by the SKA telescope (grey) will dwarf today's data sets in genomics and climate science.

Encyclopedia of DNA Elements (ENCODE), 2012
15 TB

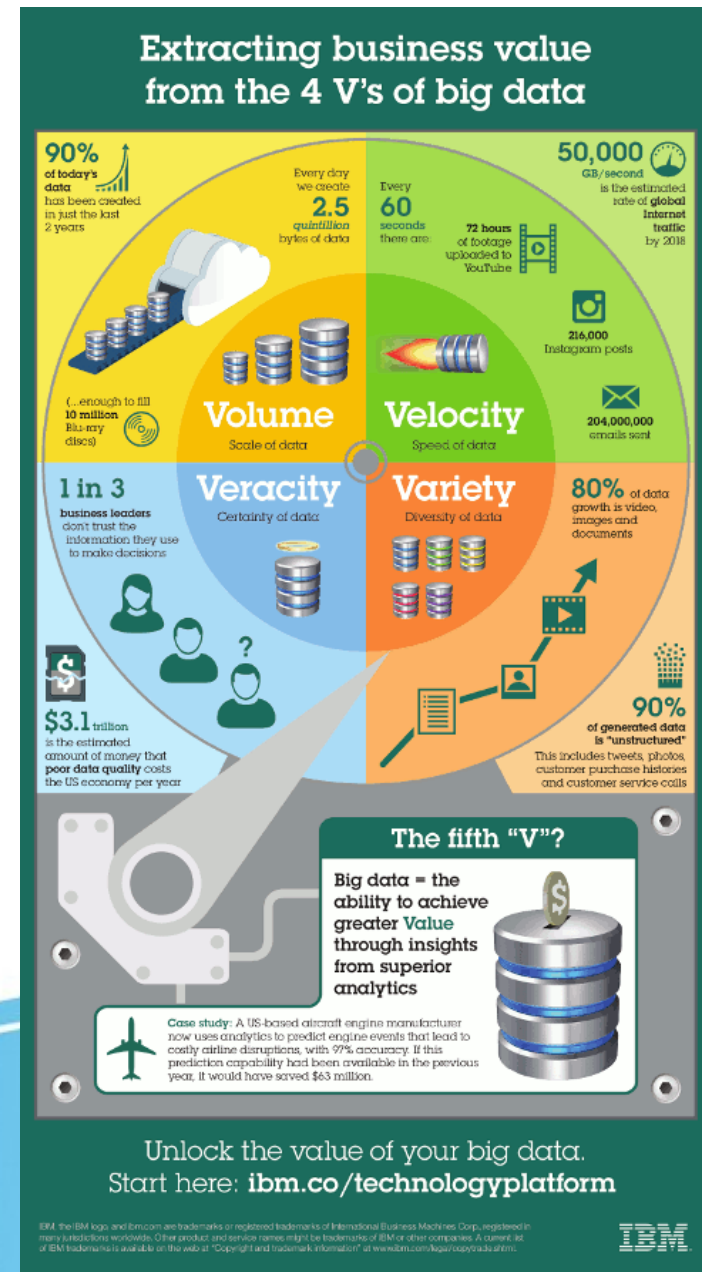
US National Climate Assessment (NASA projects), 2013
1,000 TB

Fifth assessment report by the Intergovernmental Panel on Climate Change (IPCC), due 2014
2,500 TB

Square Kilometre Array (SKA), first light due 2020
22,000,000,000 TB per year

In a nutshell: the 3 → 4 V's

- ***“Big data is high-volume, high-velocity and/or high-variety information assets that demand cost-effective, innovative forms of information processing that enable enhanced insight, decision making, and process automation.”***
- Source: Gartner at <http://www.gartner.com/it-glossary/big-data>
- Don't forget about veracity
 - And validity and volatility ...



Source: IBM's Big Data & Analysis Hub

Challenges

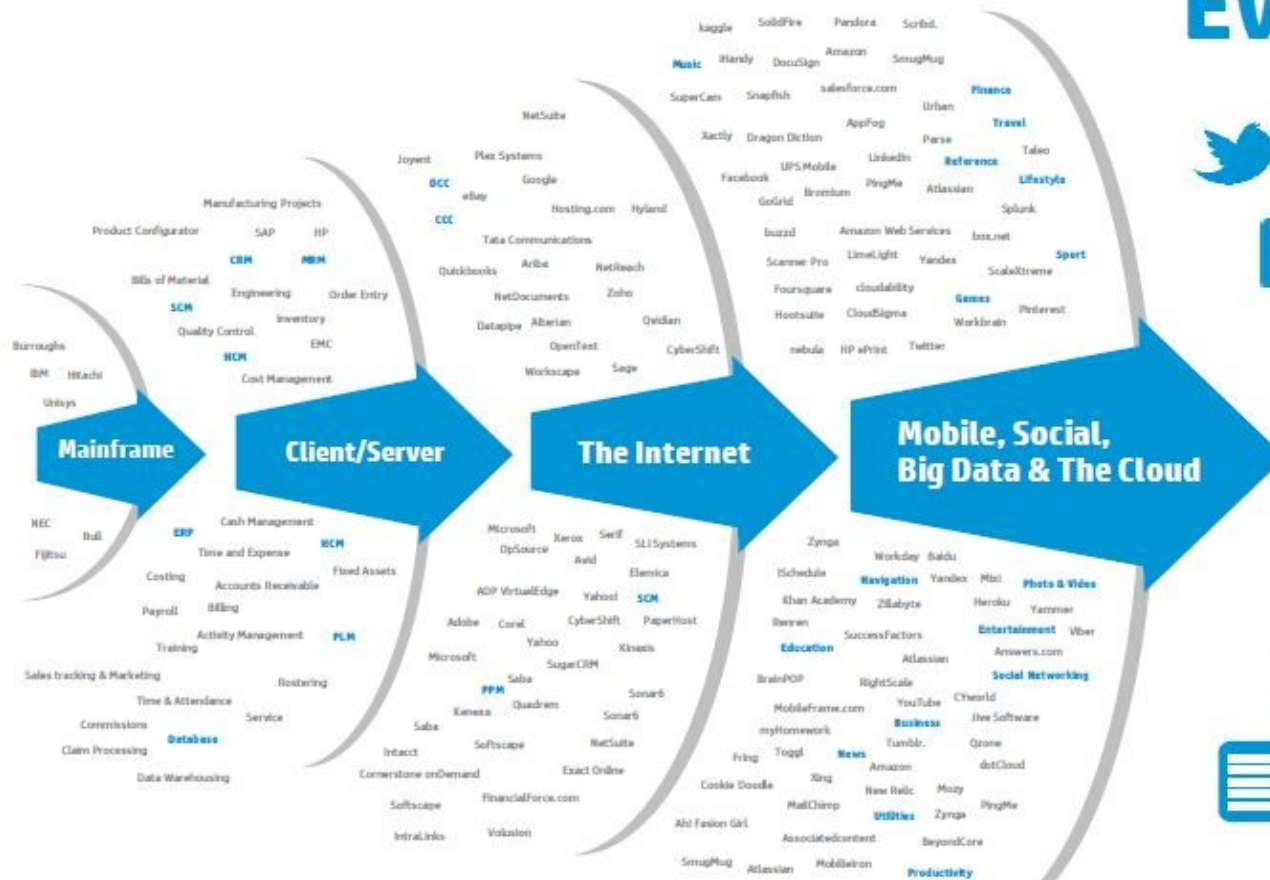


Volume

- Your data do not fit in main memory
 - A search engine query-log
 - A snapshot of the Web
 - The user pairwise-similarity matrix in a collaborative filtering system
 - ...
- Your data do not fit on secondary storage on a single machine
 - A Web corpus
 - Large Hadron Collider's data (~100 GB/s)
 - <http://home.web.cern.ch/about/computing>

Velocity

A new style of IT emerging



Every 60 seconds



98,000+ tweets



695,000 status updates



11 million instant messages



698,445 Google searches



168 million+ emails sent



1,820TB of data created



217 new mobile web users

<https://www.internetlivestats.com/>

Variety



- Heterogeneous sources
- Structured vs unstructured (often unstructured or semi-structured)
- Part of this data is transient (e.g., streamed videos)
- Only part of it is available for analysis

Issues: a random list

- Data may not be stored permanently
 - Only 1% of LHC data is analyzed (on [sciencealert](#))
- Moving data is expensive
- It may be impossible to keep all data in a single place
- In general, space becomes an issue
- Distributed/parallel programming
- Heterogeneous programming environments
- Performance is main issue
 - A normally trivial computation may prove infeasible
 - e.g., assume you have the feature vectors for 1 million users and you want to compute pairwise similarities. Assume computing similarity between 2 users requires 10 ns (we assume computation is performed in main memory). How long does your computation require using the naïve method?

A look at the data

- Text (possibly with some structure)
 - e.g., Web pages
- Graphs
- Sequences
 - e.g., search engines or recommender systems' logs, human genome
- Data often stored as simple text files with more or less standard formats (e.g., csv)

```
2::1207::4::978298478
2::1968::2::978298881
2::3678::3::978299250
2::1244::3::978299143
2::356::5::978299686
2::1245::2::978299200
2::1246::5::978299418
2::3893::1::978299535
2::1247::5::978298652
3::3421::4::978298147
3::1641::2::978298430
3::648::3::978297867
3::1394::4::978298147
3::3534::3::978297068
```

Applications domains



Some typical tasks

- Retrieving items related to a given query (query may be implicit)
 - Search engines
 - Continuous, DB-like queries
 - E.g.: top-k most interesting trends
 - Number of distinct IP flows
 - Classification
 - E.g., which Web pages are spam?
- Recommending items
 - Products (e.g., Amazon)
 - Documents of potential interest (e.g., Google alerts)
 - New links (e.g., new friendships on FB)
 - Recommending pages for personalized ads
- Detecting group of items that are related in some way
 - e.g., communities in social networks, similar Web pages

Seems little but actually a lot! Many applications fit one of the above high level descriptions, which are themselves related

Caveats and traps



When is “big” really big?

- It is a result of your data size and goals vs your computational resources
 - Data volume vs memory
 - Computational/communication complexity vs available resources
 - A moving target
 - What is big now may not be in the future
 - Some things are going to stay ...
 - If your algorithm requires $\Omega(2^n)$ steps too often
 - Is it the algorithm?
 - Is it the problem?
- 

Data collection traps/1

- Assume we have a political election in country C with two parties A and B
 - Assume the ideal case in which we collect a sample of Twitter (now X) users (representative of the totality of country C's X users), 60% of which like party A
 - What can we say about the accuracy of this predictor of the election's outcome?
 - Elaborate on the factors affecting the accuracy of our estimate
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- A decorative blue wavy bar at the bottom of the slide, consisting of two overlapping curved shapes in different shades of blue.

Data collection traps/2

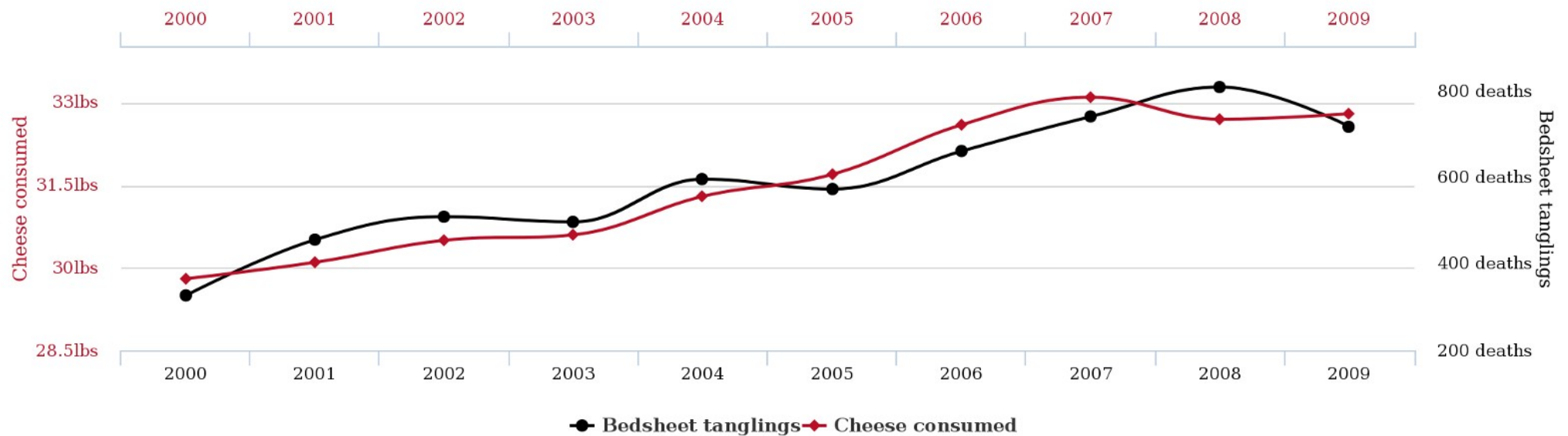
- Data are the result of a sampling process → potential bias
 - Filtering: some attributes may be omitted → loss of possibly important information, introduction of artificial correlations/removal of real ones
 - At least part of context is missing
- 

Data analysis traps/1

Per capita cheese consumption

correlates with

Number of people who died by becoming tangled in their bedsheets



tylervigen.com

<http://www.tylervigen.com/spurious-correlations>
Remember Bonferroni's principle :-)

Data analysis traps/2: Bonferroni's principle

- Data can always present some unusual patterns that look significant but are not
- e.g., a DB table where rows are people and columns are attributes. You may have thousands of attributes and thus make hypotheses that seem to be supported by data only by chance → a few funny examples further in this course (other examples are less funny)

“...if you just grab a few of them [hypotheses] ... because the data seem to suggest that—there’s some chance you will get lucky. The data will provide some support. But unless you’re actually doing the full-scale engineering statistical analysis to provide some error bars and quantify the errors, it’s gambling.” [IEEE Spectrum's interview with Michael Jordan](#)

Data interpretation traps/1

- Sleeping with one's shoes on is strongly correlated with waking up with a headache.
- *Therefore, sleeping with one's shoes on causes headache*

Interpretations traps/2

- Causality vs correlation: does $A \rightarrow B$ or are they simply correlated?
 - Check
https://en.wikipedia.org/wiki/Correlation_does_not_imply_causation
 - Generality: Is the population the data refer to representative of a larger population?
 - We are mostly working on biased samples
 - Information-rich attributes may have been filtered out during data cleaning
 - A lot of context is missing
 - e.g.: using social networking platform data to infer trends in society may be misleading if not done carefully
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Wrapping up

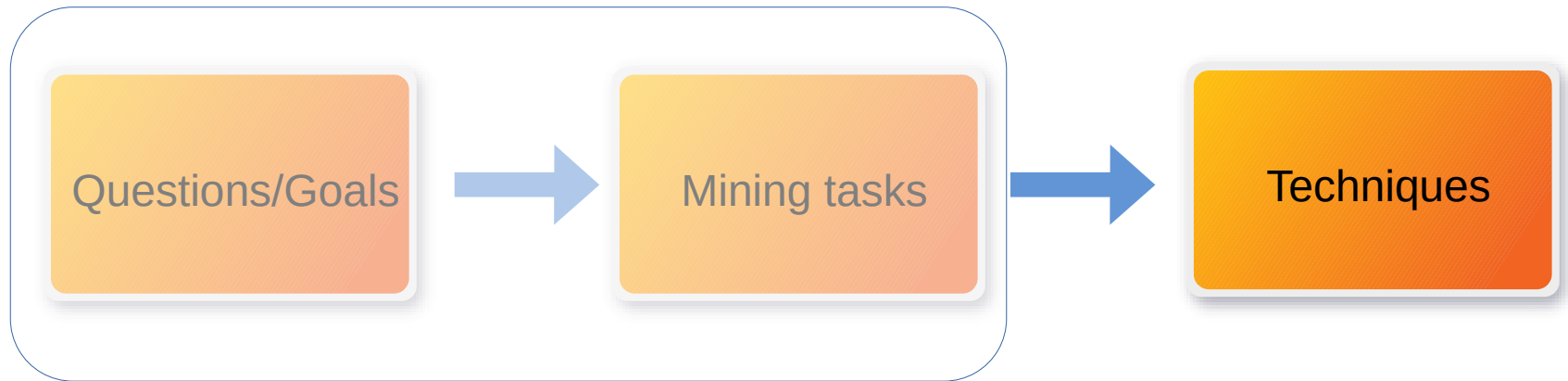
- Need a lot of knowledge and critical thinking to analyze data
 - Data are “projections” of often complex processes
 - Part of the context is lost in collection or is simply unavailable for collection
 - Maybe a wealth of information in the data, but need tools to tell statistically significant patterns from bogus



“Data don’t make any sense,
we will have to resort to statistics.”

Course outline

Course philosophy



Prerequisites

- Calculus and basic knowledge of probability theory and statistics
 - Programming, fundamental algorithms and data structures
 - Some linear algebra (we cover what needed)
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Logistics

- Register to the mailing list: <http://aris.me/registerDM.html>
 - Web page: [**http://aris.me**](http://aris.me)
 - Class hours
 - Physical attendance (no remote attendance , sorry)
 - Office hours: send email
 - Google classroom: **pzsdkxdd**
 - Books, notes
 - Exam
 - Hackathon
 - Written exam
 - Class participation
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Topics

- Similarity measures
- Text mining
- Clustering
- Neighborhood search
- Data modeling
- The curse of dimensionality and dimensionality reduction

What should I study?

- Try to take as much as you can from classes
- We have a main reference (the book on massive datasets) + references expanding on specific topics or covering topics non discussed in the book
- Spend time on references at home, trying to understand key notions
 - Discuss with friends/colleagues
 - Ask instructor
- Studying on slides is very poor practice
 - Slides **are not** studying material
 - Necessarily incomplete
 - Important details missing
 - Without instructor's comment, some claims may even be inaccurate

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